

Problem Set I

Q1. (i) Each equation has one root. Use Newton's Method to approximate the root to eight correct decimal places. (a) $x^3 = 2x + 2$ (b) $e^x + x = 7$ (c) $e^x + \sin x = 4$.

(ii) Choosing an appropriate interval or initial guesses, find the roots of the above equations using the bisection and secant method to eight decimal places.

Q2. Apply Newton's Method to find the only root to as much accuracy as possible, and find the root's multiplicity. Then use Modified Newton's Method to converge to the root quadratically.

(a) $27x^3 + 54x^2 + 36x + 8$ (b) $36x^4 - 12x^3 + 37x^2 - 12x + 1$

Q3. (i) Use Lagrange interpolation to find a polynomial that passes through the points. (a) (0,1), (2,3), (3,0) (b) (-1,0), (2,1), (3,1), (5,2)

(ii) Use Newton's divided differences to find the interpolating polynomials of the points in (i), and verify agreement with the Lagrange interpolating polynomial. Show plots of the interpolating polynomial functions.

Q4. Let $P(x)$ be the degree 5 polynomial that takes the value 10 at $x = 1, 2, 3, 4, 5$ and the value 15 at $x = 6$. Find $P(7)$.

Q5. Give an example of the following, or explain why no such example exists. (a) A degree 6 polynomial $L(x)$ that is zero at $x = 1, 2, 3, 4, 5, 6$ and equal to 10 at $x = 7$. (b) A degree 6 polynomial $L(x)$ that is zero at $x = 1, 2, 3, 4, 5, 6$, equal to 10 at $x = 7$, and equal to 70 at $x = 8$.

Q6. (a) Use the method of divided differences to find the degree 4 interpolating polynomial $P_4(x)$ for the data (0.6, 1.433329), (0.7, 1.632316), (0.8, 1.896481), (0.9, 2.247908), and (1.0, 2.718282).

(b) Calculate $P_4(0.82)$ and $P_4(0.98)$. (c) The preceding data come from the function $f(x) = e^{x^2}$. Use the interpolation error formula to find upper bounds for the error at $x = 0.82$ and $x = 0.98$, and compare the bounds with the actual error. (d) Plot the actual interpolation error $P(x) - e^{x^2}$ on the intervals $[.5, 1]$ and $[0, 2]$.

Q7. (i) The total world oil production in millions of barrels per day is shown in the table that follows. Determine and plot the degree 9 polynomial through the data. Use it to estimate 2010 oil production. Does the Runge phenomenon occur in this example? In your opinion, is the interpolating polynomial a good model of the data? Explain.

Year	Bbl/day (x1000000)
1994	67.052
1995	68.008
1996	69.803
1997	72.024
1998	73.400
1999	72.063
2000	74.669
2001	74.487
2002	74.065

2003	76.777
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(ii) Use the degree 3 polynomial through the first four data points to estimate the 1998 world oil production. Is the Runge phenomenon present?

Q8. Find the equations and plot the cubic spline that interpolates the data points (a) (0,3), (1,5), (2,4), (3,1) (b) (-1,3), (0,5), (3,1), (4,1), (5,1).